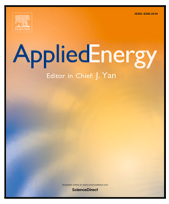
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Modeling seasonal thermal storage dynamics in the year-round scheduling of renewable energy systems☆

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ABSTRACT

Seasonal thermal storage (STS) enables the long-term storage of renewable energy in heat, which could effectively address the seasonal mismatch between renewable energy supply and heat demand. This paper models the temperature distribution of the water-based STS, considering the insulating effects of the soil surrounding the tank. Considering that the commonly used state-of-charge (SOC) model could not describe the time-variant heat loss in detail in a renewable energy system scheduling problem, a temperature field correction method is proposed to correct the scheduling results. Three case studies are performed on Garver's 6-node system and HRP-38 system to validate the proposed method's enhanced accuracy in managing STS compared to the SOC model. The proposed method could also reduce renewable curtailment in the scheduling problem due to a more detailed description of the heat loss process and a more effective renewable energy system scheduling scheme.

1. Introduction

1.1. Motivation & literature review

Renewable integration has been growing rapidly to reduce carbon emissions in energy systems. However, the increasing renewable integration results in a seasonal imbalance in the energy supply [1]. Taking China as an example, wind resources tend to be abundant in spring and autumn, while solar resources are most abundant in summer [2]. In contrast, heat demand peaks in winter, demonstrating seasonal variations due to the significant temperature differences between different seasons over the year [3]. Therefore, it is necessary to mitigate the seasonal imbalance between renewable energy and heat supply.

Seasonal thermal storage (STS), which could be used to realize long-term storage of renewable energy in heat, is a promising technology to mitigate seasonal imbalance and supply the seasonal heat load demand [4]. Traditional short-term (daily) thermal storage mainly considers the energy balance of the storage inventory within one day; In comparison, the inventory of seasonal thermal storage is assumed to be balanced within a year. There are lots of industrial applications of STS projects in renewable energy systems worldwide: In 1980, Sweden was the first country in the world to deploy STS technologies [5]; while Denmark and Germany become leading countries currently in applying STS technologies [6]. In China, demonstration projects have been applied in different provinces such as Hebei and Inner Mongolia to realize the renewable heating supply of the local energy system [7]. Currently, sensible heat storage is the most widely used STS type, which could be used to store heat in a medium such as water or soil for commercial implementation [8]. Since water could be obtained easily at relatively low economic costs [9,10], water-based STS technology is currently the most popular way to realize seasonal heat storage and will play a significant role in the future renewable-dominated power system. However, the primary barrier for STS application in the renewable energy system is the large storage volume required, leading to high investment costs with relatively limited location places for STS installation. Moreover, heat loss could not be neglected due to the long-term storage of heat. Thus, modeling the dynamics of water-based STS in renewable energy system scheduling is required for future analysis.

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Nomenclature

Superscript and subscripts

k	Index of the water layer
t	Index of time steps
w/s	Index of water/soil part
p/y	Index of proximate/distant soil part, where the soil part $s \in \{p, y\}$
$e/x/c$	Index of side/bottom/corner soil part, where the proximate soil part $p \in \{e, x, c\}$
g/r	Index of thermal/renewable generation units
l	Index of transmission lines
n	Index of bus nodes
b/h	Index of battery/seasonal thermal storage

Variables

$q_{k,t}^{\mathrm{adj}}/q_{k,t}^{\mathrm{ope}}/q_{k,t}^{\mathrm{loss}}$	Heat absorbed from the neighboring layers, the charging/discharging heat flow, and heat dissipated into the environment of the kth water layer at time t
$T_{w,k,t}/T_{s,e,t}$	The temperature of the kth layer in the water part w/side soil e in the soil part s at time t.
$m_t^{\mathrm{cha}}/m_t^{\mathrm{dis}}$	The rates of mass flow for charging/discharging water per second.
$p_t^{\mathrm{g}}/p_t^{\mathrm{r}}$	The output vector matrix of the thermal/renewable generation units.
$p_t^{\mathrm{b,cha}}/p_t^{\mathrm{b,dis}}$	The vector matrix of the charging/discharging power of battery storage
$p_t^{\mathrm{h,cha}}$	The vector matrix of the charging power of seasonal thermal storage
$f_{l,t}$	The vector matrix of the power flow
$p_{b,t}^{\mathrm{b,cha}}/p_{b,t}^{\mathrm{b,dis}}$	Charging/discharging power of the battery storage b at time t
$q_{h,t}^{\mathrm{h,cha}}/q_{h,t}^{\mathrm{h,dis}}$	Charging/discharging heat of the seasonal thermal storage h at time t
$q_{b,t}^{\mathrm{b}}/q_{h,t}^{\mathrm{h}}$	Inventory of the battery/seasonal thermal storage b/h at time t
d_t^{cur}	The curtailed load electricity.

Greek symbols

ρ_w	Density of the water part w
$\eta_h^{\mathrm{h,cha}}/\eta_h^{\mathrm{h,dis}}$	The charging/discharging efficiency of the heat exchanger connected to the seasonal thermal storage h
σ_h^{loss}	The heat loss coefficient of the seasonal thermal storage h.
$\lambda_{h,t}^{\mathrm{loss}}$	The equivalent timevariant heat loss coefficient of the seasonal thermal storage h.
$\delta_i^{\mathrm{u,rel}}/\delta_i^{\mathrm{v,rel}}$	Relative error of the energy conversion/storage calculation results.
$\delta_i^{\mathrm{u,abs}}/\delta_i^{\mathrm{v,abs}}$	Absolute error of the energy conversion/storage calculation results.

Parameters

$A_{w,x}/A_{w,k,t}$	The top/bottom and the lateral area of the kth water layer
c_s	Specific heat of the soil part s.
$U_e/U_a/U_x$	The U-values for the tank side, top and bottom of the seasonal thermal storage.
$\Delta d_{p\to d}$	Center-to-center distance between soil part p and the neighboring part in the direction d.
$c^{\mathrm{g}}/c^{\mathrm{cur}}$	The variable fuel cost vector matrix and the unit punishment cost for the load curtailment.

Abbreviations

STS	Seasonal Thermal Storage
SOC	State of Charge
HRP	High Renewable Penetration
USTS	Unit Seasonal Thermal Storage
MILP	Mixed Integer Linear Programming
COP	Coefficient of Performance

Sets

$\mathcal{G}$	Sets of thermal generation units
$\mathcal{R}$	Sets of renewable generation units
$\mathcal{H}$	Sets of seasonal thermal storage devices
$\mathcal{B}$	Sets of battery storage devices
$\mathcal{L}$	Sets of transmission lines

Many studies have considered STS in the renewable energy system optimization problem. From the perspective of renewable energy system operation and planning, the modeling of STS is usually based on the commonly used state-of-charge (SOC) model [11,12], which only focuses on tracking changes in the storage inventory. Most studies extend the energy balance constraints of the SOC model to the whole year without describing the dynamic operation characteristics of STS in the renewable energy system optimization [13,14]. However, it is essential to note that initially designed for battery storage modeling, the SOC model fails to capture the detailed dynamic changes of the STS tank's temperature field and mass flow.

Moreover, the detailed modeling of STS has also been widely discussed in many studies. Differential equations have been applied to describe the temperature field in the water storage tank so that the charging/discharging behavior could be described in detail [15]. Thermal stratification is introduced in [16] in modeling the water tank, including the structural designs of storage tanks and matched heat exchangers. Six stratified temperature-distributed models with different numerical computation complexities are formulated in [17]. Vertical stratified and stepped temperature distribution models are developed in [17,18], which are commonly used for operation simulation considering the energy balance constraints and the energy charging/discharging behavior. A vertical stratified stepped temperature distribution model is presented in [17] to describe the temperature change of the water tank. The temperature change in the surrounding environment is discussed in [19], considering the insulation effect of the soil around the storage tank. A soil-temperature model is proposed, giving a simulation result for the thermal water storage tank over several days. Most of the studies mentioned above describe the operation of STS in a non-linear physical form without linearization [20,21]. Table 1 compares different studies on seasonal thermal storage in different energy systems.

From the perspective of energy system optimization, it is crucial to note that the current STS formulation, based on the SOC model, may not be precise enough to capture the dynamic changes in the STS inventory, particularly when factoring in the heat loss of the water tank. This highlights the need for a more accurate STS formulation to

Table 1
Comparisons of different studies.

Reference	Loss	Medium	Insulation	Modeling target	Simulation scale	Research contents
[22]	Variant	Water–LiBr	N/A	Storage tank	Storage system	Modeling absorption seasonal thermal storage cycle
[23]	N/A	Water	Soil	• Collector field • Heat exchanger • Storage tank	District heating system	Modeling dynamics of the collector field; proposing the charging/discharging control logic for the storage tank
[24]	Constant	Water	Soil	• Solar collector • Storage tank • Heat exchanger	District heating system	Exploring thermal diffusivity under soil properties of different places
[25]	Constant	Water	N/A	• Heat pump • Storage tank	Integrated electricity and heat system	Modeling the seasonal and short-term thermal storage in the integrated energy system planning
[26]	Constant	Water	Aquifer	• Boiler/chiller • Storage tank	Regional building system	Investigating the benefits of seasonal thermal storage in the regional building system

optimize the scheduling of renewable energy systems effectively. Since heat is typically stored in the STS for extended periods, sometimes even months, it is imperative to recognize the heat loss. This loss should be meticulously modeled, as it can significantly affect the performance of the renewable energy system. The heat loss of the STS is usually calculated via a heat loss coefficient in the SOC model. However, the heat loss for STS is directly determined by the temperature gradient [27] and varies with the change of the temperature field inside water tanks [28]. Therefore, a constant heat loss coefficient in the SOC model might not precisely describe the heat loss for the whole tank, resulting in an imprecise estimation of the storage inventory. This is especially true when the heat is stored in hot water for a long duration (for weeks or months). As a result, the error in estimating the storage inventory directly influences the accuracy of renewable energy system operation results.

However, from the perspective of STS modeling, the present formulated STS model, considering the heat loss in detail, is pretty complex for numerical computation in renewable energy system optimization. The formulated STS model could not be directly embedded in the renewable energy system optimization problem, posing significant challenges to analyzing the operation of STS in the renewable energy system.

Therefore, it is significant to propose a more accurate analytical model of STS compared to the widely used SOC formulation to describe the dynamic temperature field changes. The proposed STS model should also be easily embedded in the energy system optimization analysis.

1.2. Contributions and innovations

This paper proposes a detailed STS model describing the dynamic changes of the temperature field within water tanks. Then, the formulated STS model is introduced in the renewable energy system scheduling problem. A correction method is developed to calculate the temperature field for STS in the scheduling problem. The three main contributions of this paper are listed as follows:

1. Proposing an analytical model calculating the time-variant heat loss of seasonal thermal storage;
2. Modeling the insulating effects of the surrounding environment in the proposed analytical model;
3. Correcting the renewable energy system scheduling results iteratively by applying the year-round hourly curve of the temperature field within the seasonal thermal storage.

1.3. Paper organization

The rest of this paper is organized as follows: Section 2 illustrates STS structure in this paper and establishes a mathematical model for temperature field calculation. Section 3 formulates the renewable energy system scheduling model as a linear programming problem considering STS. Section 4 performs three case studies on the modified Garver's 6-node system and HRP-38 system. Finally, Section 5 concludes the paper.

2. Modeling of seasonal thermal storage

In this section, the STS model is formulated considering the heat changes between different water layers and the surrounding soils. The time-variant heat loss process is described in detail. Section 2.1 explains the physical structure of the STS. Section 2.2 describes the dynamic heat change between each part of the STS.

2.1. Physical structure

Considering that the water tank of STS has different shapes, such as cylinders and cuboids, this paper employs a cylinders-shaped water tank as the modeling object, one of the most commonly used types. The mathematical formulation proposed in this paper is not limited to a specific tank shape. It could be further applied to other tank shapes, offering new possibilities and avenues for research in STS. Moreover, the water tank discussed in this paper is made of concrete [29]. Fig. 1 depicts the STS structure in detail. The structure of the STS includes three essential parts: On the top of the water is the insulated floating cover [30]. The central part is the water (w) within the STS, serving as the medium for thermal storage; meanwhile, another part is the soil part (s) around the STS, absorbing the dissipated heat from the water tank.

In the water part (w), the temperature gradient is established with hot water residing at the upper portion and cold water accumulating toward the bottom of the tank [16]. The water temperature distribution is directly determined by the stored energy in the STS. Therefore, the water part is divided into N layers from top to bottom, as shown in the figure, The temperature of each layer is different, but the temperature for one layer is the same.

Similar to the soil model outlined in [19], the environment surrounding the storage tank could be categorized into the surrounding soil (s) and the above airspace. Considering the vertical temperature distribution within the water tank and different contact areas between the water tank's sides, top and bottom surfaces, and surrounding soil, the heat exchange process between the water tank and the soil in different directions is different. Therefore, the surrounding soil is divided into different parts to model the heat transfer process in detail. In our discussions, the surrounding soil (s) composes of four parts,

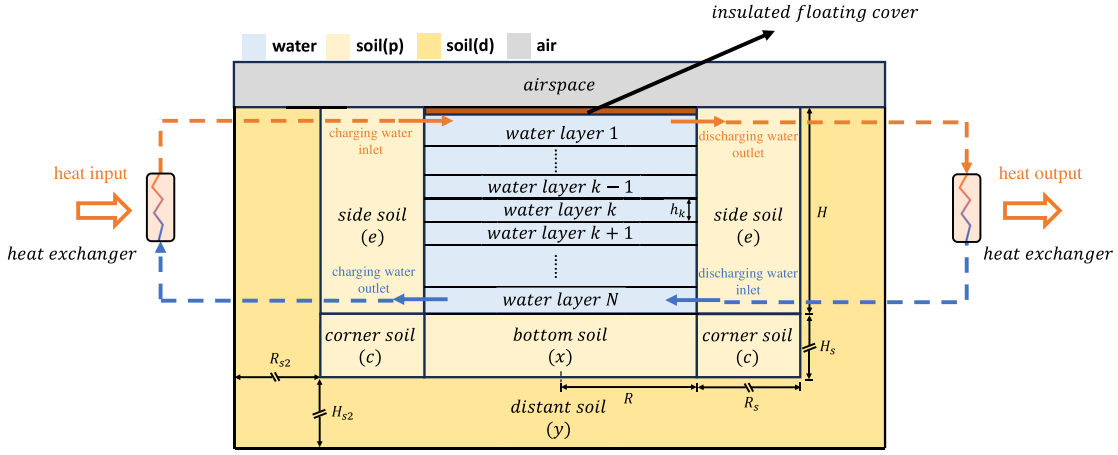


Fig. 1. Structure of the seasonal thermal storage.

the proximate soil (p) and the distant soil (y), that is $s \in \{p, y\}$. The proximate soil (p) refers to the soil that could obtain heat from the tank, while the distant soil (y) refers to the soil that only obtains heat from nearby soil. In detail, the proximate section (p) could be further divided into three distinct regions: the side soil (e), the bottom soil (x), and the corner soil (c) of the tank, that is $p \in \{e, x, c\}$. In the figure, R represents the tank's radius; H corresponds to the tank height, while H_k represents the height of the kth water layer within the tank. As for the proximate soil, R_s and H_s refer the radius and the height respectively. In addition, R_{s2} and H_{s2} represent the height and radius of the distant soil. In our discussions, it is assumed that $R_{s2} = R_s = 2.5R$ and $H_{s2} = H_s = 2.5H$ [19].

During the natural heat dissipation, the stored heat is transferred from the water to the surrounding soil, reducing the temperature differential between the water and the neighboring soil. Since the temperature differential determines heat dissipation, the surrounding soil could effectively serve as a thermal barrier, reducing the water tank's heat loss. The chilled water at the tank's base is directed out during the charging process. It then flows into the top of the tank after being heated by a properly matched electric heat pump (EHP) and the heat exchanger. In contrast, in the discharging process, the top hot water is discharged from the tank and then directed into the tank's bottom section after supplying the heat demand.

2.2. Mathematical formulation

Apart from the STS structure explained above, this paper makes the following assumptions:

1. The water part is categorized into N layers. It is assumed that the water could only flow into or out from the top first layer or the bottom Nth layer [31].
2. The mass of water flowing into the tank equals the mass flowing out from the tank simultaneously [29].
3. The temperature of the airspace part is assumed to be consistent with the ambient temperature [30].
4. The temperature of the distant soil (d) is assumed to be equivalent to the prescribed environmental temperature [32].

2.2.1. Water part

Based on the above assumptions, Section 2.2.1 first formulates the water part model. Eq. (1) formulates the energy balance for the kth water layer ($k \in [1, N]$).

$$\rho_w c_w V_{w,k} \left(T_{w,k,t+1} - T_{w,k,t}\right) = q_{k,t}^{\text{adj}} + q_{k,t}^{\text{ope}} + q_{k,t}^{\text{loss}} \tag{1}$$

where the subscript w refers to the water part, ρ_w and c_w are the density and specific heat of water, respectively. $V_{w,k}$ refers to the volume of

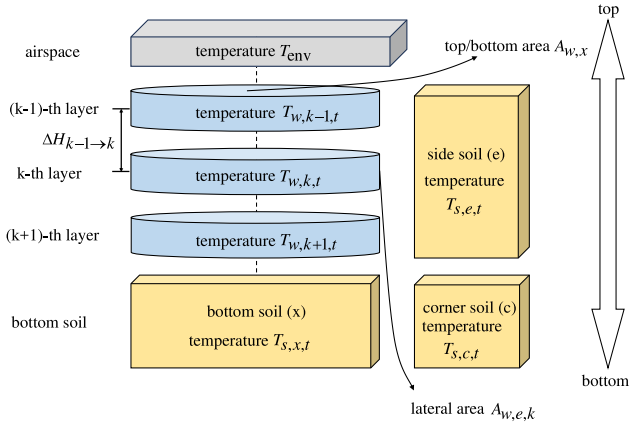


Fig. 2. Heat changing process of water layer part.

the kth layer; $T_{w,k,t}$ is the temperature of the kth layer in the water (w) part at time t. The terms $q_{k,t}^{\text{adj}}$, $q_{k,t}^{\text{ope}}$ and $q_{k,t}^{\text{loss}}$ represent the heat absorbed from the neighboring layers, the charging/discharging heat flow, and heat dissipated into the environment of the kth water layer at time t, respectively.

The heat absorbed by the kth water layer could be further calculated in detail. As shown in Fig. 2, Eqs. (2)–(4) calculate $q_{k,t}^{\text{adj}}$, $q_{k,t}^{\text{ope}}$ and $q_{k,t}^{\text{loss}}$ for the kth water layer, respectively.

$$\begin{aligned} q_{k,t}^{\text{adj}} =& \frac{k_w A_{w,x}}{\Delta H_{k-1\to k}} \left(T_{w,k-1,t} - T_{w,k,t}\right) + \\ & \frac{k_w A_{w,x}}{\Delta H_{k+1\to k}} \left(T_{w,k+1,t} - T_{w,k,t}\right) \end{aligned} \tag{2}$$

$$\begin{aligned} q_{k,t}^{\text{ope}} =& c_w m_t^{\text{cha}} \left(T_{w,k-1,t} - T_{w,k,t}\right) + \\ & c_w m_t^{\text{dis}} \left(T_{w,k+1,t} - T_{w,k,t}\right) \end{aligned} \tag{3}$$

$$\begin{aligned} q_{k,t}^{\text{loss}} =& U_e A_{w,e,k} \left(T_{s,e,t} - T_{w,k,t}\right) + \\ & x_1 \cdot U_a A_{w,x} \left(T_{\text{env}} - T_{w,1,t}\right) + \\ & x_N \cdot U_x A_{w,x} \left(T_{s,x,t} - T_{w,N,t}\right) \end{aligned} \tag{4}$$

$$x_1 = \begin{cases} 1, k = 1 \\ 0, k \in [2, N] \end{cases}, x_N = \begin{cases} 1, k = N \\ 0, k \in [1, N-1] \end{cases} \tag{5}$$

where k_w represents the thermal conductivity of water; T_{env} stands for the specified environmental temperature. $A_{w,x}$, $A_{w,e,k}$ are the top/bottom and the lateral area of the kth water layer, respectively; $\Delta H_{k+1\to k}$ and $\Delta H_{k-1\to k}$ is the center-to-center distance from the kth water layer

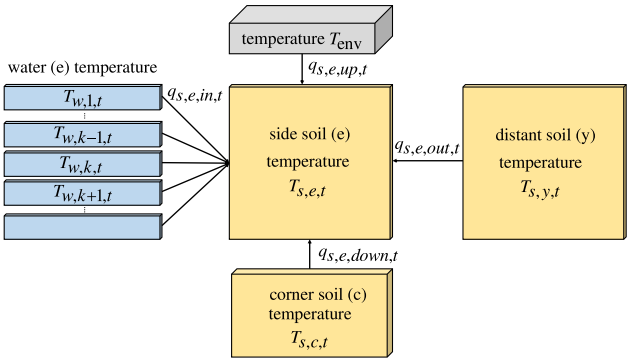


Fig. 3. Heat transfer process of soil part: taking side soil as an example.

to the neighboring layers; m_t^{cha}/m_t^{dis} are the rates of mass flow for charging/discharging water per second; U_e, U_a and U_x are the coefficient of thermal conductivity (U-values) for the tank side, top cover and bottom, respectively [33].

Particularly, the formulations for the top first and the bottom Nth water layers should be modified in Eqs. (2)–(4) as follows:

1. In Eq. (2), since there is no upper layer of the top first layer and no lower layer of the bottom Nth layer, the first term should be deleted for $k = 1$, while the second term should be deleted for $k = N$.
2. In Eq. (3), $T_{w,k-1,t}$ should be replaced with T_{cha} for $k = 1$, and $T_{w,k+1,t}$ should be replaced with T_{dis} for $k = N$, describing the charging and discharging process for the STS. T_{cha} and T_{dis} are both constants, representing the prescribed water temperature after heating and supplying the heat load respectively.
3. As shown in Eqs. (4)–(5), two binary variables (x_1 and x_N) are introduced to describe the heat loss from the top and bottom water layer of the tank, respectively. The values of x_1 and x_N equal to 1 only when the constraint (4) refers to the top and the bottom layer respectively, otherwise the value of x_1 and x_N are both zero for other layers.

2.2.2. Soil part

Fig. 3 depicts the heat transfer process of the proximate soil part (p). The models of the proximate soil (p part) around the STS are formulated as follows: Eq. (6) describes the heat changing process of the proximate soil p (including the side (e), corner (c) and bottom (x) part).

$$\rho_s c_s V_{s,p}\left(T_{s,p,t+1} - T_{s,p,t}\right) = \sum_{d\in D} q_{s,p,d,t}, p \in \{e, c, x\} \tag{6}$$

where ρ_s and c_s are the density and specific heat of the soil, respectively; $V_{s,p}$ refers to the volume of the proximate soil part p; $T_{s,p,t}$ refers to the temperature of the soil part p at time t; The set D includes four directions, namely, upward (up), downward (down), inward (in), and outward (out). $q_{s,p,d,t}$ indicates the heat received by the proximate soil part p from direction d at time t.

In detail, $q_{s,p,d,t}$ is formulated in Eq. (7):

$$q_{s,p,d,t} = \frac{k_s A_{s,p,d}}{\Delta d_{p\to d}}\left(T_{s,p,d,t} - T_{s,p,t}\right) \tag{7}$$

where k_s represents the thermal conductivity of the soil; $A_{s,p,d}$ and $\Delta d_{p\to d}$ is the area and center-to-center distance between soil part p and the neighboring part in the direction d, respectively; $T_{s,p,d,t}$ is the temperature of the part in the direction d of soil part p at time t. While if the temperature of the soil part $T_{s,p,d,t}$ is lower than the neighboring part $T_{s,p,t}$, the value of $q_{s,p,d,t}$ is negative, indicating the heat is actually flowing out in this direction.

Specifically, the equations for side soil part e in the inward direction (in) and bottom soil part x in the upward direction (up) should be modified. The side and bottom soil are in direct contact with the water tank from these directions. The revised equations are presented as follows:

$$q_{s,e,in,t} = U_e \sum_{k=1}^{N} A_{w,e,k}\left(T_{w,k,t} - T_{s,e,t}\right) \tag{8}$$

$$q_{s,x,up,t} = U_x A_{w,x}\left(T_{w,N,t} - T_{s,x,t}\right) \tag{9}$$

Eqs. (8) and (9) calculate the heat absorbed by the side soil $q_{s,e,in,t}$ and bottom soil $q_{s,b,up,t}$ dissipated from the water part, respectively. The heat loss of STS is calculated in Eq. (10) based on Eq. (4):

$$q_t^{loss} = \sum_{k=1}^{N} q_{k,t}^{loss} \tag{10}$$

where q_t^{loss} signifies the total heat loss for the STS at time t.

3. Renewable energy system scheduling considering seasonal thermal storage

As shown in Fig. 4, the scheduling of renewable energy systems with STS needs to consider producing and delivering electricity and heat. This paper explores the potential of employing a heat pump with STS to supply the heating demand. This section incorporates the STS model proposed in Section 2 into the renewable energy system scheduling model. In the proposed STS model, the charging/discharging water mass flow m_t^{cha}/m_t^{dis} and the temperature of each water layer $T_{w,k,t}$ are both variables, formulating a product expression as shown in Eq. (3). The product expression in the model significantly increases the calculation burden. To effectively tackle the scheduling problem involving STS, we have developed an iterative calculation method. This method efficiently decouples the two variables in the product expression, ensuring a high level of efficiency and practicality in the solution process. The calculation process is systematically structured into three steps, as illustrated in Fig. 5.

As shown in the figure, the iterative details are further explained as follows:

Step I The renewable energy system scheduling problem is solved considering STS based on the commonly used SOC model. In Step I, STS's year-round hourly charging/discharging behavior could be determined.

Step II Based on STS's calculated charging/discharging behavior in Step I, the year-round hourly curve of the temperature field within the STS is simulated. STS's calculated charging/discharging curve in Step I is applied as the boundary information for simulating the temperature field within STS.

Step III The scheduling results in Step I are further corrected based on the year-round hourly curve of the temperature field calculated in Step II accordingly, enabling the decoupling of the temperature field simulation and charging/discharging water flow calculation.

3.1. Renewable energy system operation simulation based on SOC model

This part formulates the scheduling problem, including the objective function, operation constraints, and STS model based on the commonly used SOC formulation. The details are as follows:

(1) Objective function

Eq. (11) shows the objective function, including the operating cost and curtailment cost:

$$\min C^{Ope} = \sum_t \left(c^g p_t^g + c^{cur} d_t^{cur}\right) \tag{11}$$

where C^{Ope} is the overall operating cost of the system; c^g and p_t^g are the variable fuel cost vector matrix and output vector matrix of the

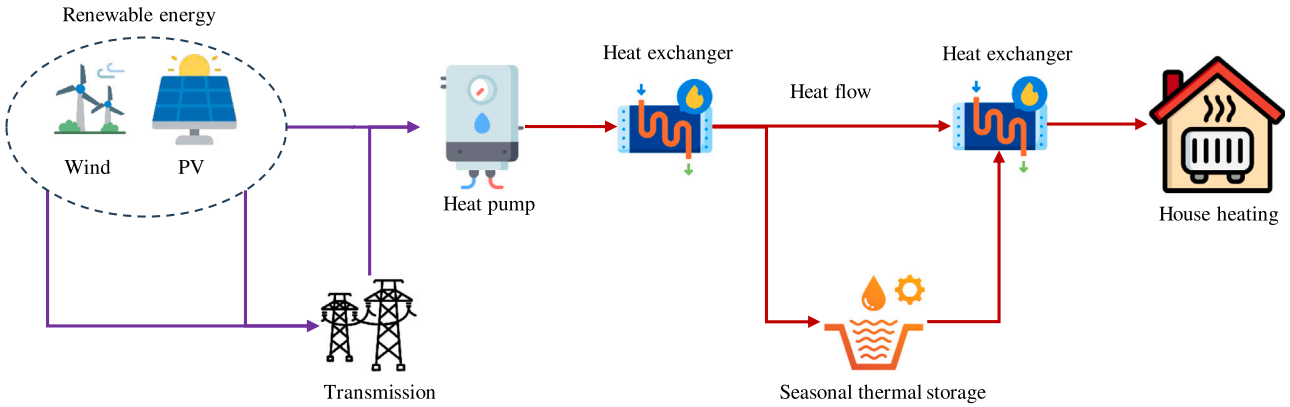


Fig. 4. Renewable energy system with seasonal thermal storage.

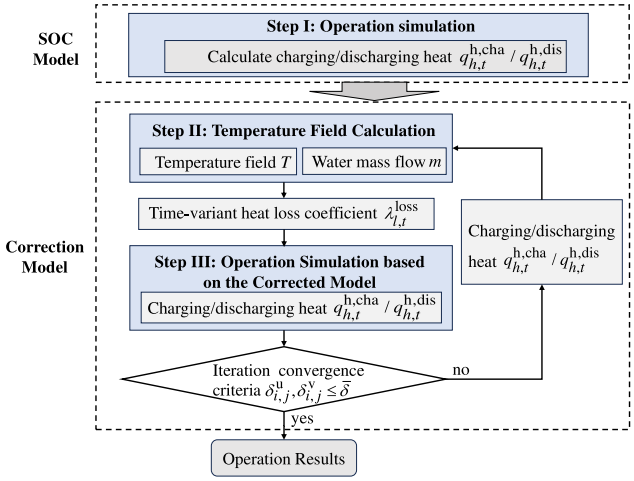


Fig. 5. Iteration process of the scheduling problem.

thermal units; c^{cur} and d_t^{cur} are the unit punishment cost for the load curtailment and the hourly curtailed load electricity.

(2) Operation constraints

$$p_t^g = \left(p_{g,t}^g\right), p_t^r = \left(p_{r,t}^r\right), \forall g \in \mathcal{G}, r \in \mathcal{R}$$
$$p_t^{b,dis} = \left(p_{b,t}^{b,dis}\right), p_t^{b,cha} = \left(p_{b,t}^{b,cha}\right), \forall b \in \mathcal{B}$$
$$p_t^{h,cha} = \left(p_{h,t}^{h,cha}\right), \forall h \in \mathcal{H} \tag{12}$$
$$d_t = \left(d_{n,t}\right), d_t^{cur} = \left(d_{n,t}^{cur}\right), \forall n \in \mathcal{N}$$
$$f_t = \left(f_{l,t}\right), \forall l \in \mathcal{L}$$

$$p_t^{gen} = \left(p_t^g, p_t^r, p_t^{b,dis} - p_t^{b,cha}, -p_t^{h,cha}\right)^T \tag{13}$$

$$p_t^{in} = p_t^{gen} + \left(d_t - d_t^{cur}\right) \tag{14}$$

$$1^T \cdot p_t^{in} = 0 \tag{15}$$

$$f_t = \pi \cdot p_t^{in} \tag{16}$$

$$0 \le p_{g,t}^g \le u_g^g \tag{17}$$

$$0 \le p_{r,t}^r \le u_r^r \omega_{r,t}^r \tag{18}$$

$$-\overline{f}_l \le f_{l,t} \le \overline{f}_l \tag{19}$$

$$0 \le d_{n,t}^{cur} \le d_{n,t} \tag{20}$$

$$\begin{cases} q_{b,t+1}^b = q_{b,t}^b + \eta_b^{b,cha} p_{b,t}^{b,cha} - p_{b,t}^{b,dis} / \eta_b^{b,dis} \\ 0 \le q_{b,t}^b \le u_b^{b,s} \\ 0 \le p_{b,t}^{b,cha}, p_{b,t}^{b,dis} \le u_b^{b,c} \\ q_{b,t=t_0}^{b,s} = q_{b,t=t_1}^{b,s} \end{cases} \tag{21}$$

Eqs. (12)–(21) formulate the operation constraints for the renewable energy system. Eq. (12) formulates the matrix vector for each variable. Meanwhile, $g/r/b/h/l$ are the subscripts of the traditional, renewables, battery, STS, and transmission lines, with $\mathcal{G}$, $\mathcal{R}$, $\mathcal{H}$, $\mathcal{B}$ and $\mathcal{L}$ serving as the sets for each device kind. Eq. (13) establishes the matrix vector p_t^{gen} for all generation/storage types. Eq. (14) calculates the input power p_t^{in} considering the power generation and load demand. Eq. (15) is the power balance constraint for the whole system. Eq. (16) calculates the power flow for the transmission lines, where π is the GSDF (Generation Shifted Distribution Factor) describing the relationship between the input power and the power flow in the system. Eq. (17)–(18) model the output of the thermal units and renewable generation units. u_g^g and u_r^r refer to the installed capacity of thermal generation and renewable generation respectively, $\omega_{r,t}^r$ is the unit renewable output curve. Eq. (19) sets the upper/lower limits for the power flow, $\overline{f}_l$ is the limit for the transmission line l. Eq. (20) limits the curtailment for the load demand. Eq. (21) establishes the operation constraints for the battery storage. The first equation describes the energy balance constraints for the battery between neighboring hours, where $\eta_b^{b,cha}/\eta_b^{b,dis}$ are the charging/discharging efficiency for the battery storage. The second and third constraints limit the inventory and the charging/discharging capacity for the battery storage. The last equation sets the energy balance constraint for the battery storage where t_0 and t_1 indicate the start and the end of each day.

(3) SOC Model for Seasonal Thermal Storage

Current studies generally establish the STS model based on the SOC formulation [11], while the STS inventory is assumed to be yearly balanced. This paper proposes the following corrections to the SOC model for STS as shown in Eqs. (22)–(26).

$$\underline{\alpha}_h^h u_h^{h,s} \le q_{h,t}^h \le \overline{\alpha}_h^h u_h^{h,s} \tag{22}$$

$$q_{h,t=0}^h = q_{h,t=8760}^h \tag{23}$$

$$q_{h,t}^{h,cha} = c_h^{hp} p_{h,t}^{h,cha} \tag{24}$$

$$q_{h,t+1}^h = \left(1 - \sigma_h^{loss}\right) q_{h,t}^h + \eta_h^{h,cha} q_{h,t}^{h,cha} - q_{h,t}^{h,dis} / \eta_h^{h,dis} \tag{25}$$

$$0 \le p_{h,t}^{\mathrm{h,cha}} \le u_h^{\mathrm{h,pc}}$$
$$0 \le q_{h,t}^{\mathrm{h,cha}}, q_{h,t}^{\mathrm{h,dis}} \le u_h^{\mathrm{h,qc}} \tag{26}$$

Eq. (22) introduces the upper/lower limits for the STS inventory: $\underline{\alpha}_h^{\mathrm{h}}/\overline{\alpha}_h^{\mathrm{h}}$ are the minimum/maximum proportions of the storage inventory, while $u_h^{\mathrm{h,s}}$ is the energy storage capacity of the STS; $q_{h,t}^{\mathrm{h}}$ refers to the stored heat in the STS h at time t. The superscript h and s indicates the storage type and capacity for STS; Eq. (23) represents the yearly energy balance constraint to ensure the yearly balance in the temperature field. Eq. (24) describes the conversion from electricity to heat via heat pump, where c_h^{hp} is the coefficient of performance (COP) of the EHP connecting to the STS. $p_{h,t}^{\mathrm{h,cha}}$ is the charging power of heat pump, while $q_{h,t}^{\mathrm{h,dis}}$ indicates the produced heat from the heat pump. Eq. (25) describes the change of storage inventory between neighboring hours: $\eta_h^{\mathrm{h,cha}}/\eta_h^{\mathrm{h,dis}}$ represent the charging/discharging efficiency of the heat exchanger connected to the STS h; σ_h^{loss} is the heat loss coefficient of the STS; Eq. (26) establishes the maximum charging and discharging heat flow limits for the heat pump and heat exchanger connected to the STS, $u_h^{\mathrm{h,pc}}$ is the energy conversion capacity of the heat pump. In contrast, $u_h^{\mathrm{h,qc}}$ is the capacity limit for the heat exchanger.

However, it is not accurate enough to describe the heat loss of the STS with a single heat loss coefficient σ_h^{loss}, especially when the temperature field inside the water tank changes during a year. Heat loss calculation would directly determine the real-time STS inventory, impacting the scheduling results. Therefore, in our following discussions, it is required to refine the scheduling results by considering the temperature field within the STS to correct the heat loss calculation in Eq. (25).

3.2. Temperature field calculation

Generally speaking, the commonly used SOC model aggregates multiple STS units within a region into one large storage tank to simplify computation, resulting in a large tank volume. However, the heat loss is closely related to the water tank volume. The aggregated STS with larger volume tends to show a lower heat loss ratio, leading to an underestimate of the heat loss of the STS storage in Section 3.1.

This paper first divides the aggregated STS into several small and identical units, referred to as unit STS (USTS). Subsequently, the volume, temperature field and heat loss for each USTS are computed separately. Finally, the heat loss is summed to represent the combined natural heat dissipation for the clustered STS units. The calculation process of the temperature field is composed of two steps in detail:

Step I Determine the radius R and height H of a single STS tank according to the optimized energy capacity.

Step II Calculate the temperature field based on the charging/discharging $q_{h,t}^{\mathrm{h,cha}}/q_{h,t}^{\mathrm{h,dis}}$ and inventory curve $q_{h,t}^{\mathrm{h}}$ of the STS in Section 3.1.

The volume, height and radius of a single USTS could be computed as follows: Eq. (27) calculates the volume of USTS, with the energy storage capacity of the STS divided by the number of the storage unit cluster $n_h^{\mathrm{h,u}}$. Eqs. (28) and (29) define the maximum height and radius constraints for a USTS, respectively.

$$V_h^{\mathrm{h,u}} = \frac{u_h^{\mathrm{h,s}}}{N \cdot \rho_w c_w \left(\overline{T} - \underline{T}\right)} \tag{27}$$

$$0 \le H_h^{\mathrm{h,u}} \le \overline{H}_h^{\mathrm{h,u}} \tag{28}$$

$$0 \le R_h^{\mathrm{h,u}} \le \overline{R}_h^{\mathrm{h,u}} \tag{29}$$

The selection of $n_h^{\mathrm{h,u}}$ should ensure that the USTS units have appropriate sizes. $V_h^{\mathrm{h,u}}$, $H_h^{\mathrm{h,u}}$ and $R_h^{\mathrm{h,u}}$ are the volume, height and radius of a single USTS, respectively. $\overline{T}$ and $\underline{T}$ are the maximum and minimum temperatures of the water within the STS, respectively; $\overline{H}_h^{\mathrm{h,u}}$ and $\overline{R}_h^{\mathrm{h,u}}$ correspond to the maximum allowable height and radius of a single USTS.

The calculation of the temperature field for a single USTS is based on Eqs. (1)–(9), where the mass of the water flow m_t^{cha} and m_t^{dis} are calculated as follows:

$$m_{h,t}^{\mathrm{h,u,cha}} = \frac{\eta_h^{\mathrm{h,cha}} q_{h,t}^{\mathrm{h,cha}}}{n_h^{\mathrm{h,u}} \cdot c_w \left(T_{\mathrm{cha}} - T_{w,N,t-1}\right)} \tag{30}$$

$$m_{h,t}^{\mathrm{h,u,dis}} = \frac{q_{h,t}^{\mathrm{h,dis}}/\eta_h^{\mathrm{h,dis}}}{n_h^{\mathrm{h,u}} \cdot c_w \left(T_{w,1,t-1} - T_{\mathrm{dis}}\right)} \tag{31}$$

$$m_{h,t}^{\mathrm{h,cha}} = n_h^{\mathrm{h,u}} m_{h,t}^{\mathrm{h,u,cha}}, m_{h,t}^{\mathrm{h,dis}} = n_h^{\mathrm{h,u}} m_{h,t}^{\mathrm{h,u,dis}} \tag{32}$$

Eqs. (30)–(32) model the charging and discharging water flows respectively. Similar to the capacity subdivision method in Eq. (27), the charging/discharging power of the STS is evenly divided into $n_h^{\mathrm{h,u}}$ parts for each USTS. $m_{h,t}^{\mathrm{h,u,cha}}/m_{h,t}^{\mathrm{h,u,dis}}$ are the charging/discharging water for a single USTS as depicted in Fig. 1. By starting with an appropriate initial value for the water temperature $T_{w,1\sim N,0}$ and soil temperature $T_{s,e/c/x,0}$, the temperature field within a USTS could be calculated step by step in a whole year.

Moreover, Eqs. (33)–(36) formulate the constraints for temperature field calculation when $k \in [1, N]$. Eq. (33) imposes limits on the water temperature within the STS. Eq. (34) limits the water flow masses during the charging and discharging processes, respectively.

$$\underline{T} \le T_{w,k,t} \le \overline{T} \tag{33}$$

$$\underline{m}_h^{\mathrm{h}}/N \le m_{h,t}^{\mathrm{h,u,cha}}, m_{h,t}^{\mathrm{h,u,dis}} \le \overline{m}_h^{\mathrm{h}}/N \tag{34}$$

where $\underline{m}_h^{\mathrm{h}}$ and $\overline{m}_h^{\mathrm{h}}$ denote the minimum and maximum allowable water flow masses for the whole STS.

The heat loss for the USTS is defined in Eq. (10). The total natural heat loss could be calculated in Eq. (35):

$$q_{h,t}^{h,\mathrm{loss}} = N \cdot q_{h,t}^{\mathrm{h,u,loss}} \tag{35}$$

where $q_{h,t}^{\mathrm{h,u,loss}}$ and $q_{h,t}^{\mathrm{h,loss}}$ represent the heat loss of a single USTS and the entire STS respectively. The equivalent time-variant heat loss coefficient $\lambda_{l,t}^{\mathrm{loss}}$ is defined in Eq. (36):

$$\lambda_{h,t}^{\mathrm{loss}} = q_{h,t}^{\mathrm{h,loss}}/q_{h,t}^{\mathrm{h}} \tag{36}$$

3.3. Correction model

Based on the calculated time-variant heat loss coefficient $\lambda_{h,t}^{\mathrm{loss}}$ in Section 3.2, the SOC model in Section 3.1 is corrected as follows: Eq. (37) calculates the charging thermal energy of the STS water tank; Eq. (38) corrects Eq. (25) by replacing the constant heat loss coefficient σ_h^{loss} with the time-variant heat loss coefficient $\lambda_{h,t}^{\mathrm{loss}}$. Eqs. (39) and (40) calculate the charging/discharging heat and water flow with the water temperature results in $T_{w,1,t}/T_{w,N,t}$ calculated in Section 3.2 serving as the boundary information. Eq. (41) sets the minimum and maximum limits for the water flow during the charging and discharging process.

$$q_{h,t}^{\mathrm{h,cha}} = c_h^{\mathrm{h}} p_{h,t}^{\mathrm{h,cha}} \tag{37}$$

$$q_{h,t+1}^{\mathrm{h}} = \left(1 - \lambda_{h,t}^{\mathrm{loss}}\right) q_{h,t}^{\mathrm{h}} + \eta_h^{\mathrm{h,cha}} q_{h,t}^{\mathrm{s,cha}} - q_{h,t}^{\mathrm{h,dis}}/\eta_h^{\mathrm{h,dis}} \tag{38}$$

$$\eta_h^{\mathrm{h,cha}} q_{h,t}^{\mathrm{h,cha}} = m_{h,t}^{\mathrm{h,cha}} \left(T_{\mathrm{cha}} - T_{w,N,t}\right) \tag{39}$$

$$q_{h,t}^{\mathrm{h,dis}}/\eta_h^{\mathrm{h,dis}} = m_{h,t}^{\mathrm{h,dis}} \left(T_{w,1,t} - T_{\mathrm{dis}}\right) \tag{40}$$

$$\underline{m}_h^{\mathrm{h}} \le m_{h,t}^{\mathrm{h,cha}}, m_{h,t}^{\mathrm{h,dis}} \le \overline{m}_h^{\mathrm{h}} \tag{41}$$

The time-variant heat loss coefficient $\lambda_{h,t}^{\text{loss}}$ reflects the influence of the dynamic temperature change on the heat loss, providing a more precise estimation for the STS heat loss in the renewable energy system scheduling problem.

Furthermore, the correction method is iterated to enhance the calculation accuracy. In Eq. (42), the vector matrix $\boldsymbol{\delta}_{i,j}$ is introduced to measure the difference between the ith and jth iteration. The equation shows that the subscript i and j refer to the calculated results of the ith and jth iteration process respectively; $\delta_{i,j}^{\text{u}}$ and $\delta_{i,j}^{\text{v}}$ measure the residual sum of squares of the storage inventory and charging/discharging results during the ith and jth iteration process respectively. As for the iteration time, $i = 0$ indicates the initial scheduling result based on the SOC model; while $i \in [1,2,3,4]$ refers to the results after i times iterations. Eq. (43) shows the difference of storage inventory and charging/discharging behavior during the ith and jth iteration. Eqs. (44)–(45) calculate the error of the storage inventory $\delta_{i,j}^{\text{u}}$ and the error of the charging/discharging results $\delta_{i,j}^{\text{v}}$ respectively, where $u_h^{\text{h,s}}$ and $u_h^{\text{h,c}}$ are the installed capacity of the heat pump and the water tank respectively. Eq. (46) shows the calculation errors $\delta_i^{\text{u,rel}}$ and $\delta_i^{\text{v,rel}}$ in adjacent iterations, comparing the results of the ith iteration and the neighboring $(i+1)$th iteration. In Eq. (47), a benchmark is selected with the STS operation simulation based on considerable numbers of water layers and iteration times, providing convincing comparison results. The calculation results depict the absolute error $\delta_i^{\text{u,abs}}/\delta_i^{\text{v,abs}}$ between the results of the ith iteration and the benchmark results. The iteration goal is set in Eq. (48), where $\bar{\delta}$ is the upper bound for the iteration gap.

$$\boldsymbol{\delta}_{i,j} = \left[\delta_{i,j}^{\text{u}}, \delta_{i,j}^{\text{v}}\right] \tag{42}$$

$$\begin{aligned} \Delta q_{h,t}^{\text{h}} &= q_{h,t}^{\text{h}}\big|_i - q_{h,t}^{\text{h}}\big|_j \\ \Delta q_{h,t}^{\text{h,cha}} &= q_{h,t}^{\text{h,cha}}\big|_i - q_{h,t}^{\text{h,cha}}\big|_j \\ \Delta q_{h,t}^{\text{h,dis}} &= q_{h,t}^{\text{h,dis}}\big|_i - q_{h,t}^{\text{h,dis}}\big|_j \end{aligned} \tag{43}$$

$$\delta_{i,j}^{\text{u}} = \sum_t \left(\Delta q_{h,t}^{\text{h}}\big|_{\{i,j\}} / u_h^{\text{h,s}}\right)^2 \tag{44}$$

$$\delta_{i,j}^{\text{v}} = \sum_t \left(\left(\Delta q_{h,t}^{\text{h,cha}} + \Delta q_{h,t}^{\text{h,dis}}\big|_{\{i,j\}}\right) / u_h^{\text{h,c}}\right)^2 \tag{45}$$

$$\delta_i^{\text{u,rel}} = \delta_{i+1,i}^{\text{u}}, \delta_i^{\text{v,rel}} = \delta_{i+1,i}^{\text{v}}, i \in \{0,1,2,3\} \tag{46}$$

$$\begin{cases} \delta_i^{\text{u,abs}} = \sum_t \left(\Delta q_{h,t}^{\text{h}}\big|_{\{i,benchmark\}} / u_h^{\text{h,s}}\right)^2 \\ \delta_i^{\text{v,abs}} = \sum_t \left(\Delta q_{h,t}^{\text{h,cha}} + \Delta q_{h,t}^{\text{h,dis}}\big|_{\{i,benchmark\}} / u_h^{\text{h,c}}\right)^2 \\ i \in \{0,1,2,3,4\} \end{cases} \tag{47}$$

$$\delta_{i,j}^{\text{u}}, \delta_{i,j}^{\text{v}} \le \bar{\delta} \tag{48}$$

Based on Eqs. (11)–(48), the renewable energy system scheduling problem is formulated as a MILP (Mixed Integer Linear Programming) model, which could be effectively solved by commercial solvers such as Gurobi.

4. Case study

In this section, three case studies are performed on the modified Garver's 6-node system and HRP-38 system based on China's Northwestern power system (HRP represents high renewable penetration [34]) to validate the proposed method. Table 2 compares different settings of the three case studies: Case I presents the benchmark results with 100 water layers and after 10000 iterations; Case II performs the case study based on the SOC model with the initial results; Case III corrects the results via the proposed model in Section 3.3. Detailed comparisons are discussed in the following parts.

Table 2
Descriptions of three case studies.

Case	Description
Case I	Benchmark results after setting the layer numbers as 100 and iteration times as 10 000
Case II	Initial results based on the commonly-used SOC model
Case III	Corrected results after final iterations based on the proposed method

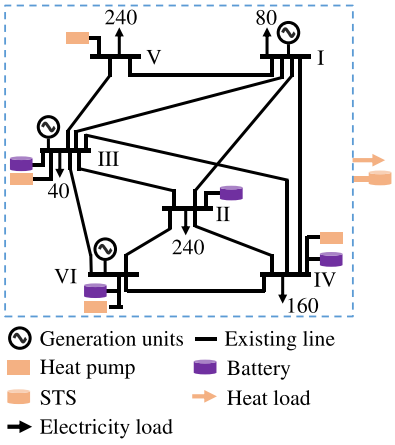


Fig. 6. Modified Garver's 6-node system.

Table 3
Detailed settings of different parameters.

Parameter	Value	Parameter	Value
c_w/c_s (J/(kg K))	4200/1600	N	15
$c^{\text{h,hp}}$	2	$U_e/U_a/U_x$	0.7/0.01/1
σ^{loss}	0.0001	k_w/k_s	0.59/3.15
ρ_w/ρ_s (kg/m^3)	1000/1400	$\eta^{\text{h,cha}}/\eta^{\text{h,dis}}$	0.75/0.75

4.1. Garver's 6-node system

4.1.1. Case settings

The proposed model is first applied to the modified Garver's 6-node system [35] as shown in Fig. 6. In the modified system, additional transmission lines connect each bus node. The battery and the heat pump are placed at the bus nodes. The heat demand of each bus node is summed together to represent the total heat demand for the power system. The STS, on the other hand, is aggregated to serve the renewable energy system.

The detailed data settings of the case studies are demonstrated in Table 3. As shown in the table, the charging/discharging efficiency $\eta^{\text{h,cha}}/\eta^{\text{h,dis}}$ of the STS are considered the same, both are 75%. The COP of the heat pump is set as 2 for simulation.

The number of water layers N is set as 15. Although an increasing number of water layers could produce a more accurate simulation result, the computation burden would grow as well. Therefore, the number of the water layers N should balance the computational efficiency and solution accuracy. This paper divides the water layer into $N = 15$ layers to obtain relatively accurate simulation results with an acceptable computation efficiency. Other detailed settings are further explained in the following discussions.

4.1.2. Results analysis

Table 4 compares the operation results under three case studies. As shown in the table, the proposed model in this paper could reduce the renewable curtailment ratio from 12% (Case II) to 8% (Case III), providing a simulation result more similar to the benchmark result (Case I). While the total heat loss accounting for the installed storage

Table 4
Comparisons of the different operation results on the Garver's 6-node power system.

Item	Case I	Case II	Case III
Thermal utilization hour (h)	4330	4356	4331
Wind utilization hour (h)	2495	2368	2497
PV utilization hour (h)	1630	1513	1631
STS heat loss (%)	41%	48%	40%
Renewable curtailment (%)	7%	12%	8%

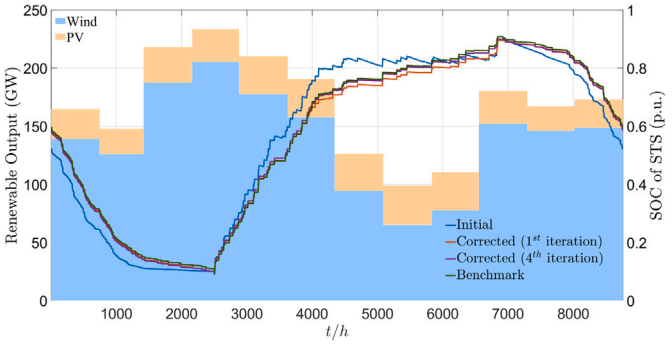


Fig. 7. SOC Curve of the STS and renewable output in the Garver's 6-node system.

capacity of STS decreases from 48% (Case II) to 40% (Case III). As a result, the utilization hours of both wind and PV are increasing. The comparisons prove that our proposed detailed STS model could more accurately describe the thermal dynamics than the commonly used SOC model, enabling the growth of renewable integration.

Fig. 7 depicts the SOC curve of the STS and the renewable output of the entire system. Comparisons between the results based on the SOC and proposed correction models are made. As the figure shows, STS mainly changes from spring to summer($t \in [3000, 4000]$), with high renewable output and no heat demand. Renewable electricity is further utilized to produce heat, which could then be stored in the STS for a long duration. During the summer–winter scenario($t \in [4000, 7000]$), natural heat dissipation decreases the STS inventory. From winter to spring ($t \in [0, 2500]$ and $t \in [7000, 8760]$), STS discharges to supply the heat load, leading to a decrease in the storage inventory.

Moreover, four curves in different colors are compared. The green curve depicts the benchmark SOC curve in Case I; the blue curve represents the initial SOC curve based on the commonly used SOC model; the orange curve represents the corrected SOC curve after one iteration; while the purple one is the corrected SOC curve after four iterations. The comparisons depict that the STS inventory is overestimated during the summer and autumn seasons ($t \in [3000, 7000]$), indicating that a constant heat loss coefficient might underestimate the heat loss during the summer–autumn season. The above results have further proven that the proposed time-variant heat loss coefficient $\lambda_{h,t}^{\text{loss}}$ could describe the dynamic heat loss process in a more detailed way.

Fig. 8 compares the values of difference $\log(\boldsymbol{\delta}_i)$ during four iterations: The iteration process converges after four times of iterations. Four indexes are introduced for comparisons: $\delta_i^{\text{u,rel}}$ and $\delta_i^{\text{v,rel}}$ are errors between the scheduling results of the $(i+1)$th and ith iterations. $\delta_i^{\text{u,abs}}$ and $\delta_i^{\text{v,abs}}$ calculate the errors between the results of the ith iteration and the final iteration.

The 0-th iteration refers to the initial scheduling result based on the SOC model, while $i \in [1, 2, 3, 4]$ refers to the results after i times iterations. As shown in Fig. 8, the relative calculation errors $\delta_i^{\text{u,rel}}$ and $\delta_i^{\text{v,rel}}$ for both the inventory curve and charging/discharging curve reduce rapidly with increasing iteration times, suggesting the results converge effectively after four iterations. Based on the above analysis, the scheduling result after four iterations ($i = 4$) is selected as the benchmark. While $\delta_i^{\text{u,abs}}$ and $\delta_i^{\text{v,abs}}$ describe the absolute difference between the results of the ith iteration and the benchmark also decreases rapidly, further proving the effectiveness of our proposed model.

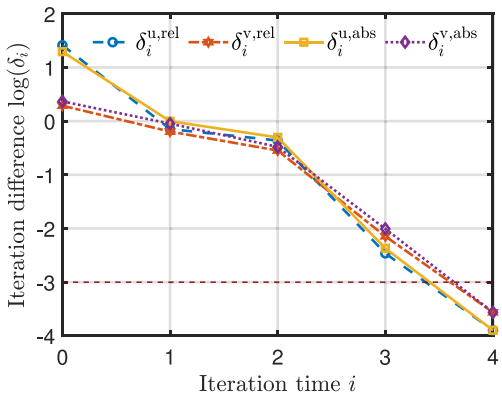


Fig. 8. Difference between the results after iterations in the Garver's 6-node system.

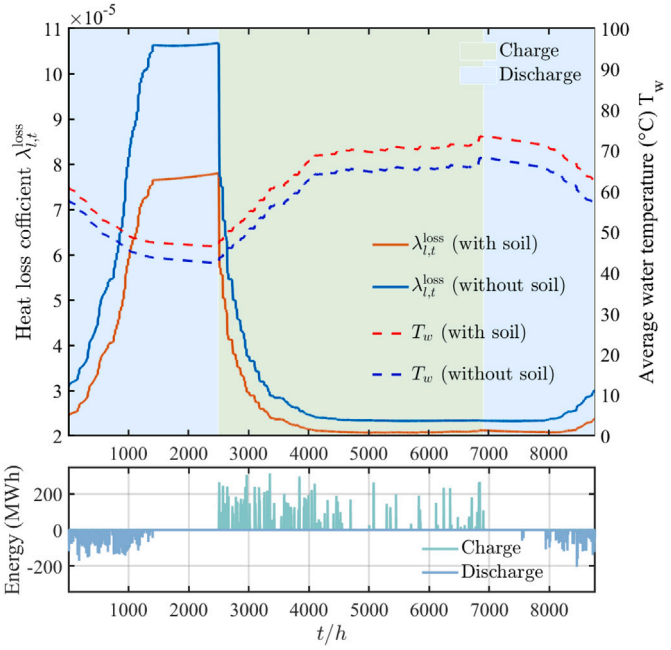


Fig. 9. Equivalent heat loss coefficient and average water temperature in the Garver's 6-node system.

Moreover, the above comparisons indicate that the commonly used SOC model might overestimate or underestimate the STS inventory during specific periods, leading to inaccurate simulation results of STS in the renewable energy system scheduling problem. This estimation error is mainly because the constant heat loss coefficient could neither reflect the heat change process between neighboring water layers in the tank nor could the coefficient describe the changing insulating effect of the surrounding soil. Therefore, the corrected results based on our proposed STS model could provide a time-variant heat loss coefficient, more accurately simulating the heat change within the water tank.

Fig. 9 displays the average water temperature and the equivalent heat loss coefficient in a year-round timescale. Two conditions are compared before and after considering the insulating effect of the surrounding soil. The top figure illustrates that considering the surrounding soil, the STS exhibits a lower heat loss with a higher average water temperature during the year. The maximum difference in the calculated heat loss coefficient is greater than 3×10^{-5}, proving that the surrounding soil could provide a good insulation performance for the water tank. Furthermore, it is evident that the time-variant heat loss coefficient $\lambda_{h,t}^{\text{loss}}$ varies across different periods and reaches the highest

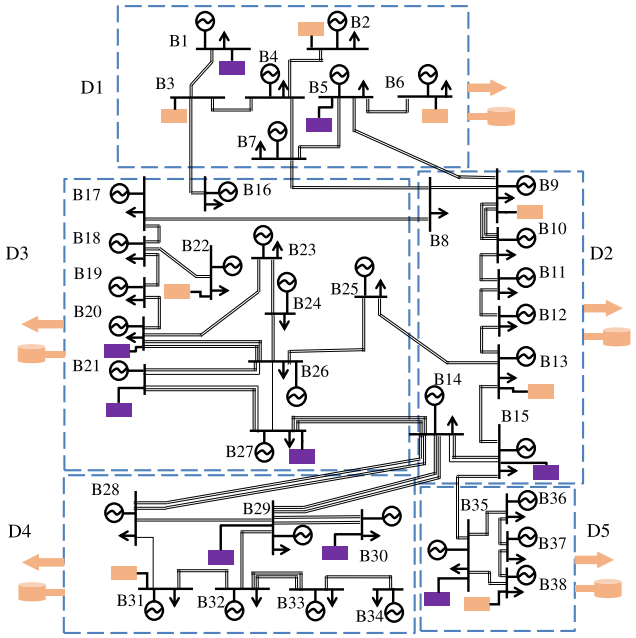


Fig. 10. Modified HRP-38 system.

Table 5
Comparisons of the different operation results on the HRP-38 System.

Item	Case I	Case II	Case III
Thermal utilization hour	4118	4189	4126
Hydro utilization hour	3211	3208	3210
Wind utilization hour	2460	2308	2466
PV utilization hour	1559	1429	1563
STS heat loss (%)	38%	43%	37%
Renewable curtailment (%)	12%	15%	11%

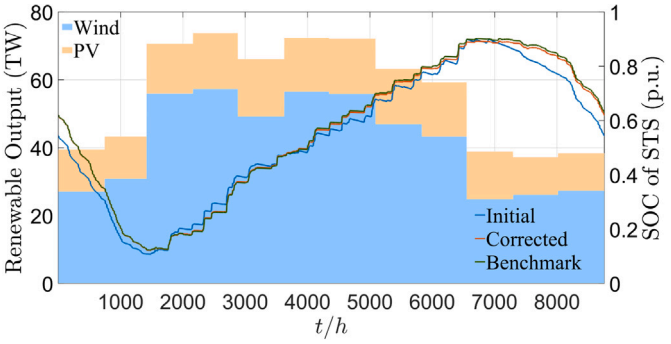


Fig. 11. STS SOC curve and renewable output in China's northwestern power system.

value when the water temperature is relatively low ($t \in [1000, 3000]$ during the winter-to-spring season).

The bottom figure describes two operation control modes in charging and discharging status, respectively. As shown in the figure, the charging of STS from renewable heating enables the increase in the average temperature of water tanks. While the discharging mode indicates the heat release to supply the heat demand, bringing down the average water temperature.

The above results have further validated that heat loss could not be accurately described by a simple constant coefficient σ_h^{loss}. The STS operation during periods with low water temperature should be incredibly detailed.

4.2. HRP-38 system

4.2.1. Case settings

Fig. 10 describes the HRP-38 system, including five regions. The system includes 38 bus nodes and 137 transmission lines with high renewable integration. The heat demand and STS within each region are aggregated. Thus, each region has one aggregated heat demand and one STS for seasonal thermal storage.

4.2.2. Results analysis

Table 5 compares the operation results based on the HRP-38 system. As shown in the table, the renewable curtailment decreased from 15% (Case II) to 11% (Case III), with the utilization hour for wind and PV increasing by more than 100 h. Based on the benchmark results in Case I, the proposed model in this paper (Case III) provides a more accurate description of the STS inventory change compared to the SOC model in Case II, thus enabling a more reasonable dispatch decision of the STS with higher renewable integration.

Fig. 11 compares the entire system's inventory curve, including the benchmark, initial results based on the SOC model and the corrected results based on our proposed model. Corrections are made based on the initial results, and the difference is relatively high during the winter season ($t \in [0, 1000]$ and $t \in [7000, 8760]$). Furthermore, we provide a detailed demonstration of the renewable output and the inventory for STS of each region in Fig. A.1, Appendix. The figure shows that the inventory difference between the initial and corrected results is great, especially for Regions I, II, and III, further verifying the necessity of remedying the inventory curve.

The STS's monthly average temperature field distribution within each region is further depicted in Fig. A.2. The volume of each water tank is the same while the total installation capacity of STS is different with various flexibility demands in each region. Each season includes three months' temperature curves. From spring to summer, the water within the STS is heating up, indicating the charging of the STS. Conversely, from winter to spring, the water within the STS cools down, accompanied by its discharge. From autumn to winter, there are only relatively slight changes in the temperature field, meaning that the hot water is stored in the tank with only heat dissipation and no charging/discharging. Based on the formulated temperature field within the STS, the energy inventory changes of the STS could be described and visualized more accurately and directly.

5. Conclusion and future works

Seasonal thermal storage is a promising solution for mitigating the seasonal mismatch between renewable supply and heat demand. It could also serve as a practical long-term storage option. This paper proposes an analytical model of seasonal thermal storage with water as the medium, The temperature distribution within the seasonal thermal storage water tank is modeled in detail while the surrounding soil serves as the thermal insulation. Moreover, this paper introduces a correction model with a time-variant heat loss coefficient describing the dynamic heat change process instead of a constant coefficient in the widely used SOC model.

Case studies on the modified Garver's 6-node system and the HRP-38 system have proven that the proposed modeling method could more accurately describe seasonal thermal storage's dynamic heat change process. In this paper, comparisons are made between the commonly used SOC model and the proposed model. The comparison results indicate that describing the heat loss in detail could contribute to more effective scheduling results for renewable energy systems and enhance renewable integration.

Future work will consider comparing different groundwater and soil types and modeling geological stratification and uneven vertical temperature distribution of soil to establish a more detailed model of seasonal thermal storage. The dynamic temperature changes and the

corresponding control strategy of the heat supply and demand side also need to be discussed within a smaller temporal resolution to investigate their impacts on the load of the heat source and heat users. Future research should also consider cross-year temperature changes in the surrounding soil.

CRediT authorship contribution statement

Haiyang Jiang: Writing – review & editing, Writing – original draft, Formal analysis, Conceptualization. **Jiajun Luo:** Visualization, Validation, Data curation. **Yan Guo:** Funding acquisition. **Ershun Du:** Supervision. **Ning Zhang:** Supervision, Data curation. **Yuchen Fang:** Visualization, Validation, Software. **Yating Wang:** Project administration. **Goran Strbac:** Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix

See Figs. A.1 and A.2.

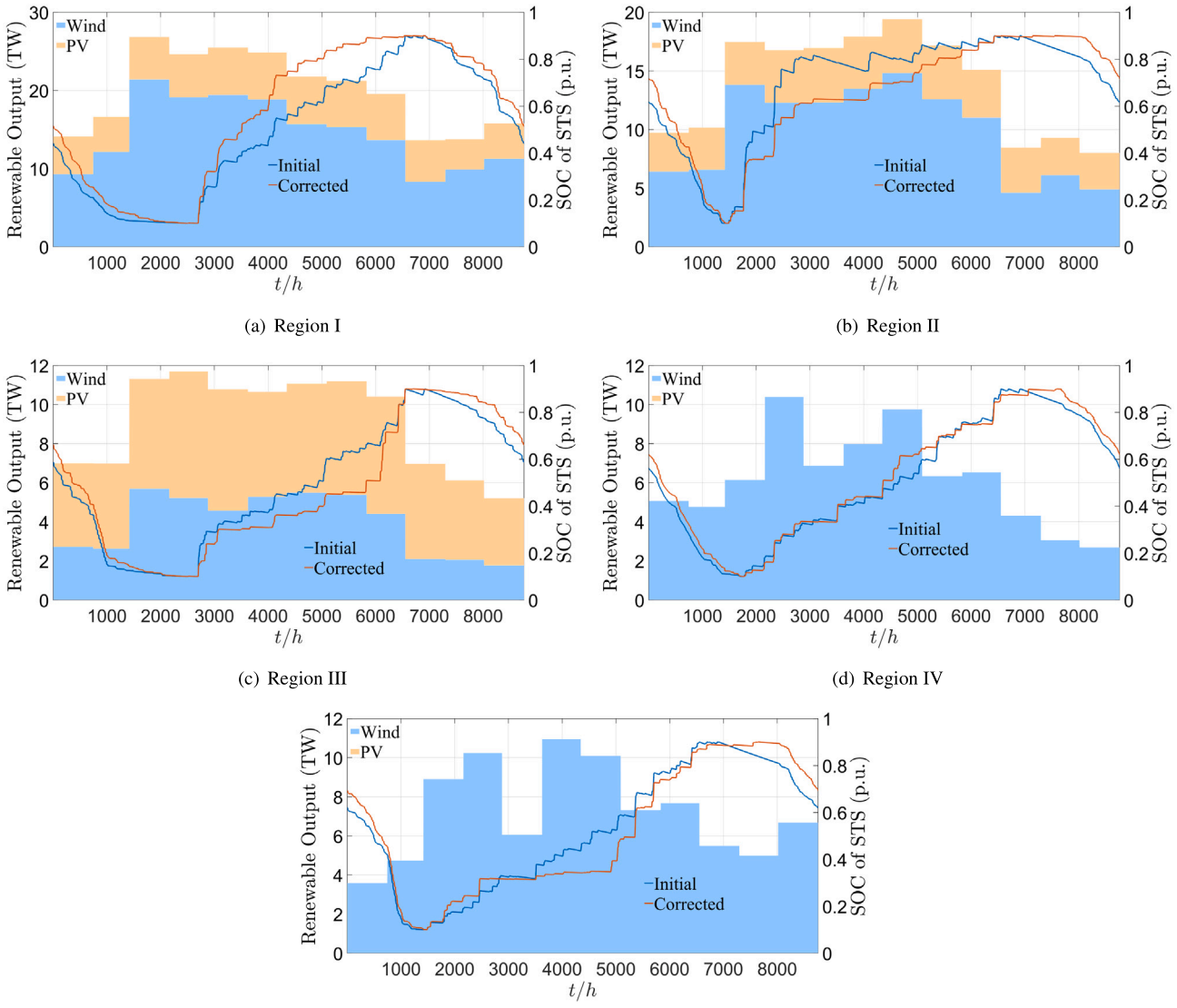
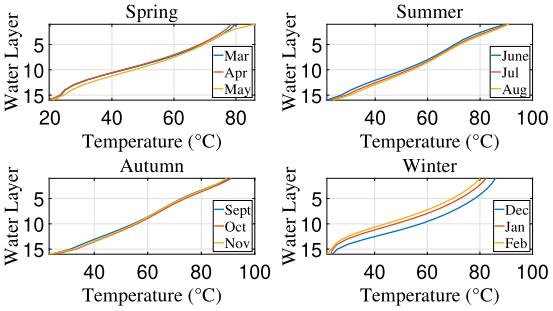
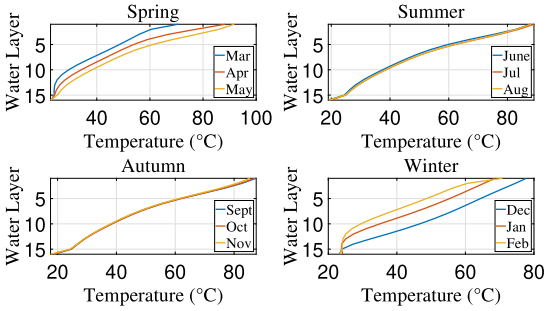


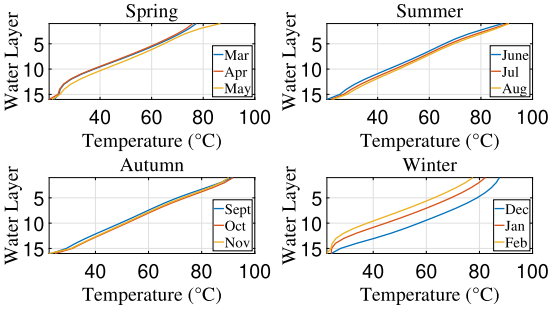
Fig. A.1. Hourly inventory changes and renewable output of each region.



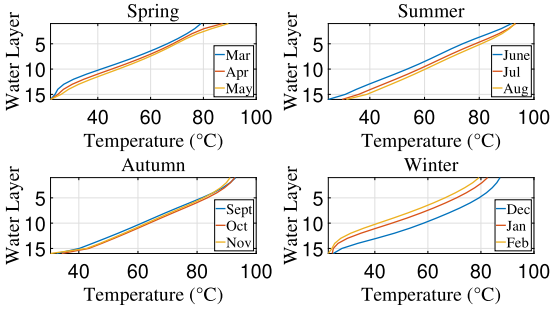
(a) Region I



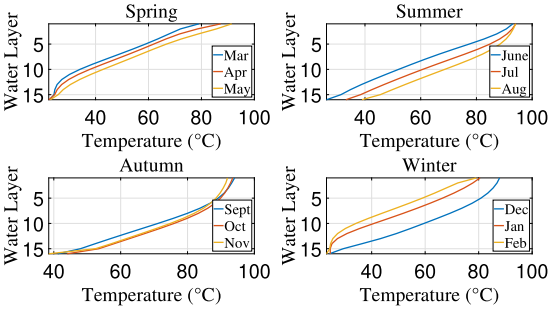
(b) Region II



(c) Region III



(d) Region IV



(e) Region V

Fig. A.2. Temperature field within the water tank for different seasons of each region.

Data availability

Data will be made available on request.

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